# Practice workbook · Plan or specify a single-factor laboratory method

Bench investigations of enzyme temperature, osmosis in potato, diffusion into agar cubes, and the detail missing from a written method.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-PLAN-CONTROLLED-LAB-METHOD

# 1. Question 5 · part a.i · 5 marks

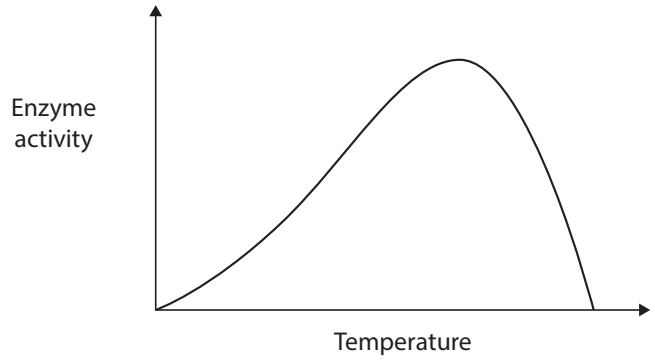
June 2024 Paper 1

Worked-example retry · Attempt part a.i only.

**5** (a) A group of students investigate the effect of temperature on enzyme activity.

The students use the results they obtain to plot a graph.

The graph shows the students' results.




(Source: adapted from https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/25-enzymes/enzyme-activity.html)

(i) Design a method the students could use to obtain the results shown in the graph.

**(5)**

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# 1. Question 5 · part a.i · 5 marks

June 2024 Paper 1

Worked-example retry · Attempt part a.i only.

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(ii) State the independent variable in this investigation.

**(1)**

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(b) Explain how enzyme activity is affected by low temperatures.

**(4)**

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**(Total for Question 5 = 10 marks)**

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## 1. Question 5 · part a.i · 5 marks · Working and answer

Working and answer

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## 2. Question 5 · part b · 5 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

**5** Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

**(2)**

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

**(5)**

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## 2. Question 5 · part b · 5 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

| | Concentration of sucrose solution in mol per dm³ | | | | |
|---|---|---|---|---|---|
| | **0.2** | **0.4** | **0.6** | **0.8** | **1.0** |
| Initial mass in g | 8.4 | 7.9 | 8.7 | 8.3 | 8.4 |
| Mass after soaking in g | 8.7 | 8.0 | 8.2 | 7.5 | 7.3 |
| Change in mass in g | + 0.3 | + 0.1 | – 0.5 | – 0.8 | – 1.1 |
| Percentage change in mass | + 3.6 | | – 5.8 | – 9.6 | – 13.1 |

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

**(2)**

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

**(2)**

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P 6 0 1 8 7 A 0 1 4 2 4

## 2. Question 5 · part b · 5 marks · Working and answer

Working and answer

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# 3. Question 3 · part b.i · 2 marks

January 2023 Paper 2

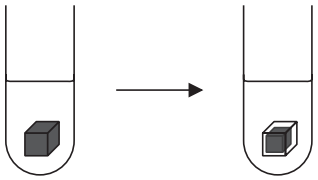
Further practice · Attempt part b.i only.

**3** A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

**(3)**

surface area to volume ratio = .....

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

**(1)**

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

**(1)**

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# 3. Question 3 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

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(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

**(2)**

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

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(i) Describe a method for this investigation using the apparatus shown in the diagram.

**(2)**

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(ii) Give two variables the student would need to control in their investigation.

**(2)**

1 .....

2 .....

**(Total for Question 3 = 11 marks)**

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## 3. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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# 4. Question 7 · part a · 4 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

**7** Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

| Time after adding enzyme in hours | Amount of lactic acid in the solution in arbitrary units | | |
|---|---|---|---|
| | 2 °C | 25 °C | 55 °C |
| 0 | 0.1 | 0.1 | 0.1 |
| 10 | 0.1 | 0.2 | 0.2 |
| 20 | 0.1 | 0.3 | 0.2 |
| 30 | 0.2 | 0.8 | 0.4 |
| 40 | 0.3 | 1.8 | 0.7 |
| 50 | 0.4 | 2.2 | 1.0 |
| 60 | 0.6 | 2.3 | 1.3 |

(a) Describe what additional information another student would need to repeat this investigation.

**(4)**

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**20**

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# 4. Question 7 · part a · 4 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

(b) (i) Estimate the optimum temperature for this enzyme.

**(1)**

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

**(2)**

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

**(3)**

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## 4. Question 7 · part a · 4 marks · Working and answer

Working and answer

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